Computational Medicine - Class 1

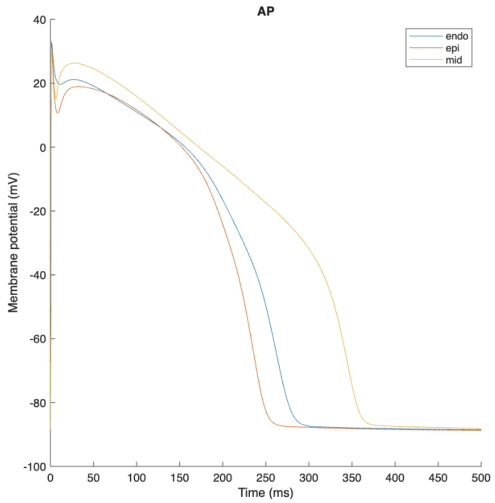
Single cell modelling and simulation

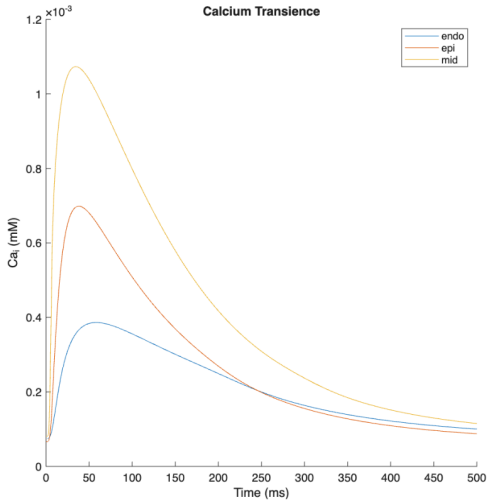
Q1. Human ToR-ORd Ventricular Cell Model

Q1a. Transmural Cellular Differences

In this section, we adjust the cell type to simulate all three ventricular cell subtypes.

The resulting action potentials and calcium transients are shown below:



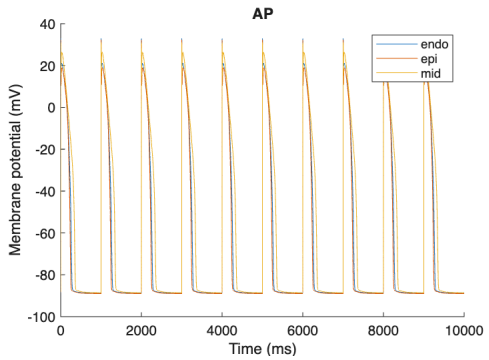


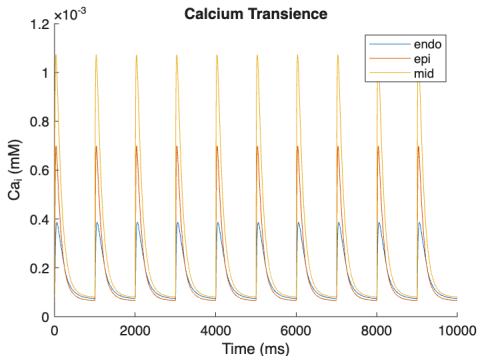
Observations:

- **Endocardial:** cells have the shortest action potential duration and the smallest calcium transient.
- **Midmyocardial:** cells exhibit the longest action potential duration and the largest calcium transient amplitude, in line with their higher ICaL conductance.
- **Epicardial:** cells show a visible early notch in the AP, most likely from a larger transient outward current.

Q1b. Visualising Several Beats

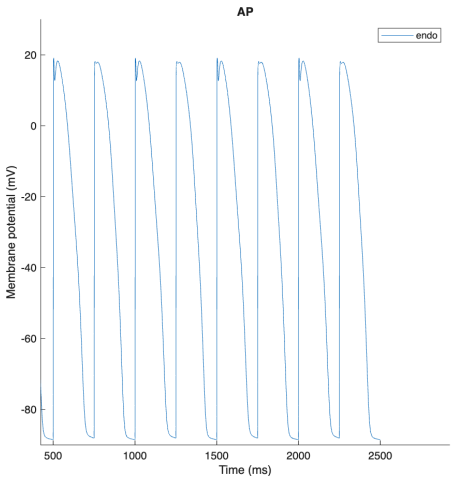
We Repeat the simulation of the previous section, but plot the last 10 beats:

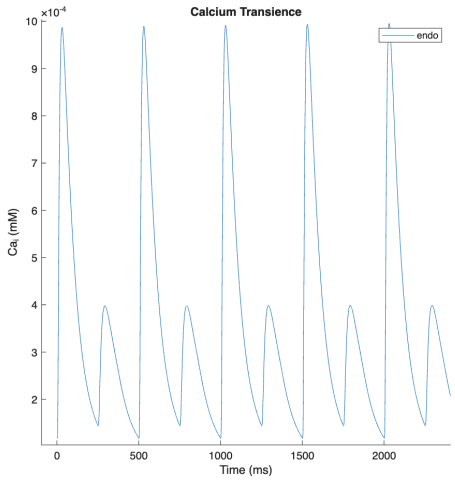




The three cell types maintain their shape across beats. The midmyocardial cell consistently shows the longest action potential duration and the highest calcium transient amplitude.

<u>Rapid Pace with BCL 250 ms and an endocardial cell</u>

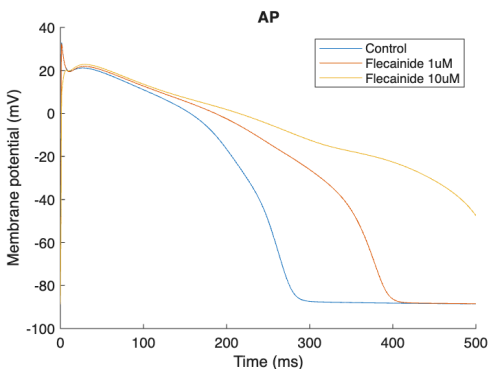


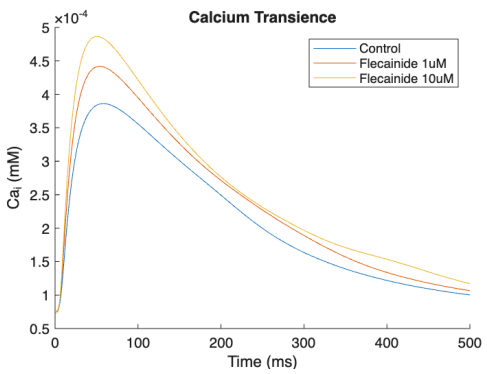


Observation: For BCL = 250 ms, the simulation alternates in both action potential duration and calcium transience amplitude.

After looking for an explanation online, I understood that this is due to the fact that at short cycle lengths, the sarcoplasmic reticulum cannot fully refill with calcium between beats.

Q2. Safety Pharmacology

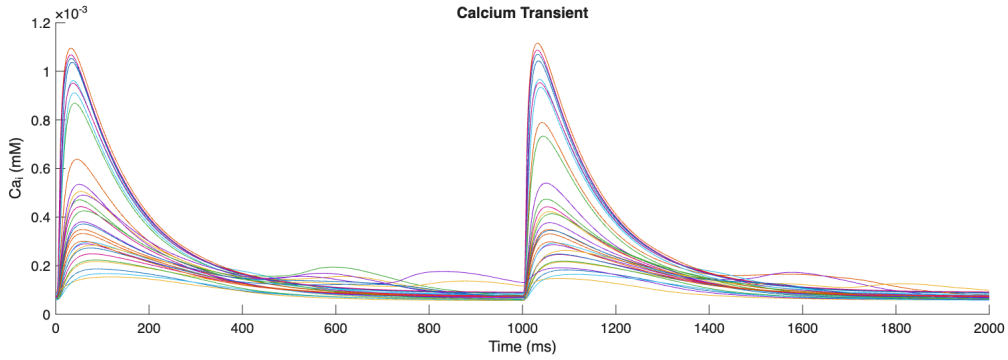




Observation

- Both 1 µM and 10 µM Flecainide produce a prolongation of action potential duration driven by the IKr coefficient. Reducing the IKr conductance indeed slows repolarisation and extends the plateau phase.
- INa has a comparatively smaller direct effect but it slightly reduces the upwards velocity and peak potential.
- Concentration has an impact on effects. 1 µM provides a 53% IKr block while 1 µM provides a 93% IKr block. The effects are prominent at the lower dose and increase slowly for higher concentrations.

Q3. Populations of Models — Flecainide at 10 µM

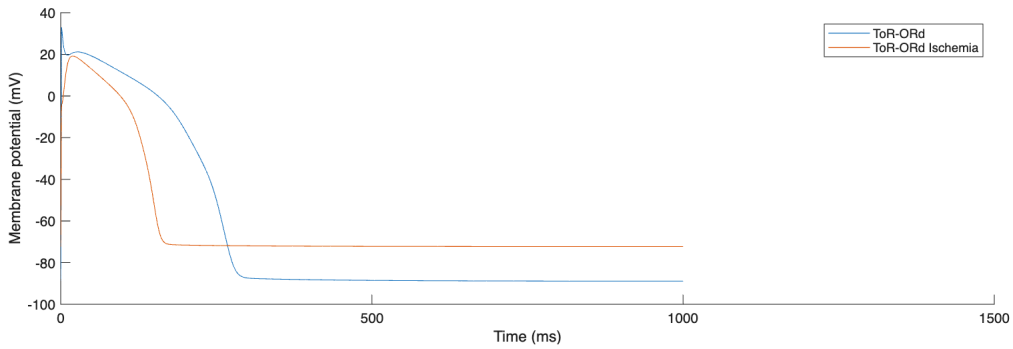


Observations:

- Some models exhibit the same behaviour as in Q1b-2, which was not observed before.
- The degree of action potential duration prolongation varies widely across the population.

The reason behind the differences observed is likely the diversity in ionic conductances. In simulations where repolarising currents are simultaneously reduced, Flecainide can create pathologies.

Q4. Ischemia Modelling



This model displays a change in action potential compared to the healthy baseline, we see a shortened action potential duration as well as a reduced peak.